

HEART FAILURE

The heart cannot pump enough blood to meet the body's needs, so blood **backs up** (congestion) and organs are **underperfused**.

Left-sided HF

L = LUNGS · backs up into the pulmonary circulation

- **Dyspnea, orthopnea** (needs more pillows), **PND** — waking breathless at night
- **Crackles** in the lungs; cough
- **Pink, frothy sputum** = pulmonary edema — emergency
- Tachycardia, **S3** heart sound
- Fatigue, weakness, activity intolerance
- Restlessness, confusion — ↓ cerebral perfusion
- ↓ Urine output — ↓ renal perfusion

Right-sided HF

R = REST OF THE BODY · backs up into systemic veins

- **JVD** — jugular vein distension
- **Dependent edema** — legs and ankles; sacrum if bedbound
- **Weight gain** from fluid retention
- **Hepatomegaly**, RUQ discomfort; **ascites**
- Anorexia, nausea, bloating — GI congestion
- Most common cause of right-sided HF: **left-sided HF**

TYPE BY EJECTION FRACTION

HFref	EF ≤ 40%	reduced — weak squeeze
HFmrEF	EF 41–49%	mildly reduced
HFpEF	EF ≥ 50%	preserved — stiff, poor filling

NYHA FUNCTIONAL CLASS

- I** No limitation of ordinary activity
- II** Slight limitation; ordinary activity causes symptoms
- III** Marked limitation; less than ordinary activity causes symptoms
- IV** Symptoms at rest

ACC/AHA STAGE

- A** At risk; no structural disease
- B** Structural disease, no symptoms (pre-HF)
- C** Structural disease with symptoms
- D** Advanced HF

DIAGNOSTICS

- **BNP / NT-proBNP** — released when ventricles stretch; elevated in HF. BNP <100 pg/mL makes HF unlikely
- **Echocardiogram** — measures EF; the key test
- **Chest x-ray** — cardiomegaly, pulmonary congestion
- **Electrolytes, renal function** — watch K⁺ and creatinine on diuretics and RAAS drugs

NURSING PRIORITIES

- **Daily weight** — same time, same scale, same clothing, after voiding, before breakfast
- **Report a gain of 2–3 lb in a day or 5 lb in a week**
- High Fowler's for dyspnea; oxygen as ordered
- Strict I&O; sodium and fluid restriction as prescribed
- Assess lung sounds, edema, JVD, and mental status every shift
- Cluster care; plan rest periods; increase activity gradually

MEDICATIONS — WHAT THE NURSE WATCHES **HFREF: 4 PILLARS ★**

Class	Examples	Nursing watch-fors
★ ARNI / ACE inhibitor / ARB	sacubitril-valsartan; -pril; -sartan	BP, hyperkalemia , renal function. ACEi: dry cough, angioedema . ARNI: never within 36 h of an ACEi
★ Beta blocker	carvedilol, metoprolol succinate, bisoprolol	Apical HR and BP before giving; hold per ordered parameters. Do not stop abruptly
★ MRA	spironolactone, eplerenone	Potassium-sparing — hyperkalemia ; renal function
★ SGLT2 inhibitor	dapagliflozin, empagliflozin	Volume depletion; genital and urinary infections. Used in HF with or without diabetes
Loop diuretic	furosemide, bumetanide	Hypokalemia , hypotension, daily weight, I&O. Give in the morning
Digoxin	digoxin	Apical pulse for a full minute; hold per parameters. Toxicity: nausea, vomiting, yellow-green halos , bradycardia. Low K⁺ raises toxicity risk . Narrow therapeutic range

⚠ **ACUTE PULMONARY EDEMA**

- Severe dyspnea, **pink frothy sputum**, crackles throughout, anxiety, hypoxia
- High Fowler's with legs dependent; oxygen
- Call the provider or rapid response
- Prepare IV loop diuretic as ordered

PATIENT TEACHING

- Weigh daily; report the gain thresholds above
- Follow the prescribed sodium limit
- **Avoid NSAIDs** — they worsen fluid retention
- Report more breathlessness, swelling, or needing more pillows to sleep

WARNING SIGNS: FACES

- Fatigue**
- Activity limitation**
- Congestion / cough**
- Edema**
- Shortness of breath**